

Answering questions with data

Mallory Barnes

2025-08-21

This comprehensive resource offers a free, accessible textbook for environmental science students embarking on introductory statistics. The package includes a practical lab manual and a dedicated course website, all provided under a CC BY-SA 4.0 license.

Table of contents

Preface	5
Acknowledgements	5
CC BY-SA 4.0 license	6
Copying the textbook	6
The vision behind this textbook	7
1 Why Statistics?	8
1.1 The Role of Statistics in Environmental Science	8
1.1.1 Why Numbers Matter	8
1.1.2 What Statistics Add	11
1.2 Introduction to measurements	11
1.3 Scales of measurement	12
1.3.1 Nominal scale	13
1.3.2 Ordinal scale	13
1.3.3 Interval scale	13
1.3.4 Ratio scale	14
1.3.5 Continuous versus discrete variables	14
1.3.6 A note on real data	15
1.4 Assessing the reliability of a measurement	15
1.5 The role of variables: predictors and outcomes	16
1.6 Experimental and non-experimental research	16
1.7 Assessing the validity of a study	17
1.7.1 Internal validity	18
1.7.2 External validity	18
1.7.3 Construct validity	18
1.8 Confounds, artifacts and other threats to validity	18
1.8.1 Fraud, deception, and self-deception	20
1.9 Summary	20
1.10 Videos	21
1.10.1 Terms of Statistics	21
2 Describing Data	22
2.1 Looking at the data	22
2.1.1 Scatter of raw values	22
2.1.2 Histograms	23

2.2	Important Ideas: Distribution, Central Tendency, and Variance	25
2.3	Measures of Central Tendency (Sameness)	25
2.3.1	Mode	25
2.3.2	Median	26
2.3.3	Mean	27
2.3.4	What does the mean mean?	28
2.3.5	Comparing the mean, median, and mode	28
2.4	Measures of Variation (Differentness)	29
2.4.1	The Range	29
2.4.2	Difference Scores	30
2.4.3	The Variance	31
2.4.4	The Standard Deviation	35
2.4.5	Mean Absolute Deviation	36
2.5	Remember to look at your data	37
2.5.1	Anscombe's Quartet	37
2.5.2	Datasaurus Dozen	38
2.6	Videos	38
2.6.1	Measures of center: Mode	38
2.6.2	Measures of center: Median and Mean	38
2.6.3	Standard deviation part I	38
2.6.4	Standard deviation part II	38
3	Correlation	39
3.1	If something caused something else to change, what would that look like? . . .	39
3.1.1	Charlie and the Chocolate factory	40
3.1.2	Scatter plots	41
3.1.3	Positive, Negative, and No-Correlation	42
3.2	Pearson's r	43
3.2.1	The idea of co-variance	44
3.3	Turning the numbers into a measure of co-variance	44
3.3.1	Co-variance, the measure	47
3.3.2	Pearson's r we there yet	49
3.4	Examples with Data	50
3.5	Regression: A mini intro	51
3.5.1	The best fit line	52
3.5.2	Lines	55
3.5.3	Computing the best fit line	56
3.6	Interpreting Correlations	58
3.6.1	Correlation does not equal causation	58
3.6.2	Correlation and Random chance	61
3.6.3	Some more movies	67
3.7	Summary	69

4	Probability, Sampling, and Estimation	70
4.1	How are probability and statistics different?	71
4.2	What does probability mean?	73
4.2.1	The frequentist view	73
4.2.2	The Bayesian view	76
4.2.3	What's the difference? And who is right?	77
4.3	Basic probability theory	77
4.3.1	Introducing probability distributions	78
4.4	The binomial distribution	84
4.4.1	Introducing the binomial	84
4.4.2	Working with the binomial distribution in R	86
4.5	The normal distribution	89
4.5.1	Probability density	92
4.6	Other useful distributions	93
4.7	Summary of Probability	94
4.8	Samples, populations and sampling	94
4.8.1	Defining a population	95
4.8.2	Simple random samples	96
4.8.3	Most samples are not simple random samples	98
4.8.4	How much does it matter if you don't have a simple random sample?	100
4.8.5	Population parameters and sample statistics	101
4.9	The law of large numbers	102
4.10	Sampling distributions and the central limit theorem	104
4.10.1	Sampling distribution of the sample means	105
4.10.2	Seeing the pieces	105
4.10.3	Sampling distributions exist for any sample statistic!	108
4.11	The central limit theorem	110
4.12	z-scores	114
4.12.1	Idea behind z-scores	115
4.12.2	Calculating z-scores	116
4.13	Estimating population parameters	118
4.13.1	Concrete population parameters	118
4.13.2	Abstract population parameters	119
4.13.3	Experiments and Population parameters	120
4.13.4	Interim summary	122
4.13.5	Estimating the population mean	122
4.13.6	Estimating the population standard deviation	123
4.14	Estimating a confidence interval	129
4.14.1	A slight mistake in the formula	130
4.15	Summary	131
4.16	Videos	131
4.16.1	Introduction to Probability	131
4.16.2	Chebychev's Theorem	131

4.16.3	Z-scores	131
4.16.4	Normal Distribution I	131
4.16.5	Normal Distribution II	131
5	Foundations for inference	132
5.1	Brief review of Experiments	132
5.2	The data came from a distribution	133
5.2.1	Uniform distribution	134
5.2.2	Not all samples are the same, they are usually quite different	135
5.2.3	Large samples are more like the distribution they came from	138
5.3	Is there a difference?	143
5.3.1	Chance can produce differences	144
5.3.2	Differences due to chance can be simulated	146
5.4	Chance makes some differences more likely than others	146
5.5	The Crump Test	150
5.5.1	Intuitive methods	151
5.5.2	Part 1: Frequency based intuition about occurrence	151
5.5.3	Part 2: Simulating chance	152
5.5.4	Part 3: Judgment and Decision-making	155
5.5.5	Part 4: Experiment Design	160
5.5.6	Part 5: I have the power	162
5.5.7	Summary of Crump Test	164
5.6	The randomization test (permutation test)	165
5.6.1	Pretend example does chewing gum improve your grades?	165
5.6.2	Take homes so far	171
5.7	Videos	171
5.7.1	Null and Alternate Hypotheses	171
5.7.2	Types of Errors	171
6	Hypothesis Testing	172
6.1	Hypothesis Testing - The Nuts & Bolts	172
6.1.1	Clarifying Alpha, P-value, and Confidence Level	172
6.1.2	The Steps of Hypothesis Testing Applied to an Example	173
6.1.3	Errors in Hypothesis Testing	174
6.1.4	Deciphering Significance with P-values	174
6.2	Graphical Review	174
6.2.1	Key Players in Hypothesis Testing Visualization	174
6.3	Graphical Review of Test Outcomes that are Not in Error	183
6.3.1	Graphical Descriptions:	183
6.4	Graphical Review of Sample Size Effect when Test Outcomes are in Error	188
6.4.1	Graphical Descriptions:	188
6.4.2	How Significant is ‘Significant’ – Interpreting p-values	190

6.5	Review of Ways to Test (H_0)	192
6.5.1	Results Statement	193
6.5.2	Beyond the 0.5 cutoff: Effect-size and power	193
6.5.3	The importance of knowing what you're doing	193
6.5.4	Chance vs. Real Effects: The Playground, the Superpower, and the Impact Scale	194
6.5.5	Effect size: concrete vs. abstract notions	195
6.5.6	Cohen's d	196
6.6	Power	198
6.6.1	A digression about hypothesis testing	198
6.6.2	Back to power	199
6.6.3	Power curves	201
6.7	Planning your design	204
6.8	Some considerations	204
6.8.1	Low powered studies	204
6.8.2	Large N and small effects	205
6.8.3	Small N and Large effects	206
6.8.4	Type I errors are convincing when N is small	206
7	t-tests	209
7.1	Check your confidence in your mean	210
7.2	One-sample t-test: A new t-test	211
7.2.1	Formulas for one-sample t-test	212
7.2.2	What does t represent?	212
7.2.3	Calculating t from data	215
7.2.4	How does t behave?	216
7.2.5	Making a decision	218
7.3	Paired-samples t-test	219
7.3.1	Mehr, Song, and Spelke (2016)	220
7.3.2	The data	221
7.3.3	The difference scores	222
7.3.4	The mean difference	223
7.3.5	Calculate t	223
7.3.6	Interpreting ts	225
7.3.7	Getting the p-values for t -values	228
7.3.8	One-tailed tests	228
7.3.9	Two-tailed tests	230
7.3.10	One or two tailed, which one?	232
7.3.11	Degrees of freedom	232
7.4	The paired samples t-test strikes back	235
7.5	Independent samples t-test: The return of the t-test?	236
7.6	Simulating data for t-tests	239
7.6.1	Simulating a one-sample t-test	239

7.6.2	Simulating a paired samples t-test	242
7.6.3	Simulating an independent samples t.test	247
7.7	Videos	247
7.7.1	One or Two tailed tests	247
8	ANOVA	249
8.1	ANOVA is Analysis of Variance	249
8.2	One-factor ANOVA	250
8.2.1	Computing the F -value	251
8.2.2	SS Total	251
8.2.3	SS Effect	253
8.2.4	SS Error	254
8.2.5	Degrees of freedom	255
8.2.6	Mean Squared Error	256
8.2.7	Calculate F	257
8.2.8	The ANOVA TABLE	257
8.3	What does F mean?	258
8.3.1	Making Decisions	260
8.3.2	Fs and means	261
8.4	ANOVA on Real Data	267
8.4.1	Tetris and bad memories	267
8.4.2	Comparing means after the ANOVA	269
8.5	ANOVA Summary	271
9	Repeated Measures ANOVA	272
9.1	Repeated measures design	272
9.2	Partitioning the Sums of Squares	273
9.3	Calculating the RM ANOVA	275
9.3.1	SS Total	278
9.3.2	SS Effect	279
9.3.3	SS Error (within-conditions)	280
9.3.4	SS Subjects	281
9.3.5	SS Error (left-over)	282
9.3.6	Check our work	282
9.3.7	Compute the MSEs	283
9.3.8	Compute F	283
9.3.9	p-value	284
9.4	Things worth knowing	284
9.4.1	Repeated vs between-subjects ANOVA	285
9.4.2	repeated measures designs are more sensitive	286
9.5	Real Data	288
9.6	Summary	290

10 Factorial ANOVA	291
10.1 Factorial basics	292
10.1.1 2x2 Designs	292
10.1.2 Factorial Notation	293
10.2 Purpose of Factorial Designs	293
10.2.1 Factorials manipulate an effect of interest	293
10.2.2 Manipulating the Pollution effect	294
10.3 Graphing the means	297
10.4 Knowing what you want to find out	297
10.5 Simplified Analysis of 2x2 Repeated Measures Design	300
10.5.1 Main Effects	300
10.5.2 Interaction	301
10.5.3 Looking at the data	301
10.5.4 Main effect of Pollution	301
10.5.5 Main effect of Flow	303
10.5.6 Interaction between Pollution and Flow	305
10.5.7 Writing it all up	307
10.5.8 2x2 Repeated Measures ANOVA	307
10.6 2x2 Between-subjects ANOVA	308
10.6.1 SS Total	308
10.6.2 SS Pollution	309
10.6.3 SS Flow	310
10.6.4 SS Pollution by Flow	310
10.6.5 SS Error	311
10.6.6 Check your work	312
10.7 Fireside chat	313
10.8 Factorial summary	313
11 More On Factorial Designs	315
11.1 Looking at main effects and interactions	315
11.1.1 2x2 designs	315
11.2 Interpreting main effects and interactions	316
11.2.1 A consistent main effect and an interaction	319
11.2.2 An inconsistent main effect and an interaction	320
11.3 Mixed Designs	321
11.4 More complicated designs	322
11.4.1 3x2 design	322
11.4.2 2x2x2 designs	325
11.4.3 Understanding and Interpreting Interactions in Environmental Science	327
12 Analysis of Covariance (ANCOVA)	329
12.1 General Linear Models (GLM)	329
12.1.1 Model Statements:	330

12.1.2	Flexibility and Applicability of GLMs	330
12.1.3	Understanding ANOVA as a linear model:	330
12.2	Moving Beyond Regression and ANOVA to ANCOVA	332
12.2.1	Understanding ANCOVA	332
12.3	Introducing the Indicator or Dummy Variable	333
12.3.1	Understanding Indicator Variables	333
12.3.2	Interaction Term in GLMs	335
12.3.3	Terminology in GLMs	336
12.4	So, What Exactly is ANCOVA?	337
12.4.1	The Null Hypothesis in ANCOVA	337
12.4.2	Testing the Null Hypothesis	338
12.5	Illustrating ANCOVA with an Example	338
12.5.1	Setting the Stage	338
12.5.2	Crafting the Model	338
12.5.3	Why We're Running the ANCOVA	339
12.5.4	Null and Alternative Hypotheses in our example ANCOVA	339
12.5.5	Analysis	340
12.5.6	Results of the ANCOVA Model	341
12.5.7	Contextualizing the Findings:	344
12.6	Chapter Summary	344
13	Thinking about answering questions with data	345
13.1	Effect-size and power	345
13.1.1	Chance vs. real effects	346
13.1.2	Effect size: concrete vs. abstract notions	346
13.1.3	Cohen's d	347
13.2	Power	350
13.2.1	A digression about hypothesis testing	350
13.2.2	Back to power	351
13.2.3	Power curves	353
13.3	Planning your design	355
13.4	Some considerations	356
13.4.1	Low powered studies	356
13.4.2	Large N and small effects	357
13.4.3	Small N and Large effects	357
13.4.4	Type I errors are convincing when N is small	358
14	GIFs	361
14.1	Correlation GIFs	361
14.1.1	N=10, both variables drawn from a uniform distribution	361
14.1.2	Correlation between random deviates from uniform distribution across four sample sizes	362

14.1.3	Correlation between random deviates from normal distribution across four sample sizes	363
14.1.4	Correlation between X and Y variables that have a true correlation as a function of sample-size	363
14.1.5	Type I errors, sampling random deviates from normal distribution with regression lines	365
14.1.6	Cell-size and correlation	366
14.1.7	Regression	367
14.2	Sampling distributions	368
14.2.1	Sampling from a uniform distribution	368
14.2.2	Sampling from uniform with line showing expected value for each number	369
14.2.3	Sampling distribution of the mean, Normal population distribution and sample histograms	369
14.2.4	Null and True effect samples and sampling means	371
14.3	Statistical Inference	372
14.3.1	Randomization Test	372
14.3.2	Independent t-test Null	373
14.3.3	Independent t-test True	376
14.3.4	T-test True sample-size	378
14.3.5	one-factor ANOVA Null	381
14.3.6	Factorial Null	383
14.4	Distributions	385
14.4.1	Normal changing mean	385
14.4.2	Normal changing sd	386

References

388

List of Figures

1.1	Crude case fatality rates. Italy appears to have a higher fatality rate than China when all cases are combined.	9
1.2	Fatality rates by age group (same made-up numbers). Within each age band, China's CFR exceeds Italy's.	10
1.3	Case age distribution by country. Italy's cases are older on average; China's are younger.	10
2.1	Pretend happiness ratings from 500 people	23
2.2	A histogram of the happiness ratings	24
2.3	Four histograms of the same data using different bin widths	24
2.4	A histogram with the mean (red), the median (green), and the mode (blue) . .	29
2.5	Values on a number line.	33
2.6	Mean as the unique balancing point. Only at the mean do signed deviations cancel.	34
2.7	Anscombe's Quartet	37
3.1	Imaginary data showing a positive correlation between amount of chocolate and amount happiness	41
3.2	Three scatterplots showing negative, positive, and zero correlation	42
3.3	Simulated sums of products showing the kinds of values than can be produced by randomly ordering the numbers in X and Y.	47
3.4	A simulation of of correlations. Each dot represents the r-value for the correlation between an X and Y variable that each contain the numbers 1 to 10 in random orders. The figure illustrates that many r-values can be obtained by this random process	50
3.5	Relationship between freedom to make life choices and confidence in national government. Data from the world happiness report for 2018	51
3.6	Three scatterplots showing negative, positive, and a random correlation (where the r-value is expected to be 0), along with the best fit regression line	52
3.7	Black dots represent data points. The blue line is the best fit regression line. The white dots are represent the predicted location of each black dot. The red lines show the error between each black dot and the regression line. The blue line is the best fit line because it minimizes the error shown by the red lines. .	53

3.8	A plot of the sum of the squared deviations for different lines moving up and down, through the best fit line. The best fit line occurs at the position that minimizes the sum of the squared deviations.	55
3.9	Two different lines with different y-intercepts (where the line crosses the y-axis), and different slopes. A positive slope makes the line go up from left to right. A negative slope makes the line go down from left to right.	56
3.10	An example regression line with confidence bands going through a few data points in a scatterplot	58
3.11	Illustration of a possible relationship between amount of water and snake plant growth. Growth goes up with water, but eventually goes back down as too much water makes snake plants die.	59
3.12	Another figure showing a range of r-values that can be obtained by chance. . .	63
3.13	A reminder of what positive, negative, and zero correlation looks like	64
3.14	A histogram showing the frequency distribution of r-values for completely random values between an X and Y variable (sample-size=10). A full range of r-values can be obtained by chance alone. Larger r-values are less common than smaller r-values	65
3.15	Four histograms showing the frequency distributions of r-values between completely random X and Y variables as a function of sample-size. The width of the distributions shrink as sample-size increases. Smaller sample-sizes are more likely to produce a wider range of r-values by chance. Larger sample-sizes always produce a narrow range of small r-values	66
4.1	An illustration of how frequentist probability works. If you flip a fair coin over and over again, the proportion of heads that you've seen eventually settles down, and converges to the true probability of 0.5. Each panel shows four different simulated experiments: in each case, we pretend we flipped a coin 1000 times, and kept track of the proportion of flips that were heads as we went along. Although none of these sequences actually ended up with an exact value of .5, if we'd extended the experiment for an infinite number of coin flips they would have.	75
4.2	A visual depiction of the pants probability distribution. There are five elementary events, corresponding to the five pairs of pants that I own. Each event has some probability of occurring: this probability is a number between 0 to 1. The sum of these probabilities is 1.	79
4.3	The binomial distribution with size parameter of $N = 20$ and an underlying success probability of $1/6$. Each vertical bar depicts the probability of one specific outcome (i.e., one possible value of X). Because this is a probability distribution, each of the probabilities must be a number between 0 and 1, and the heights of the bars must sum to 1 as well.	85

4.4	Two binomial distributions, involving a scenario in which I'm flipping a fair coin, so the underlying success probability is $1/2$. In panel (a), we assume I'm flipping the coin $N = 20$ times. In panel (b) we assume that the coin is flipped $N = 100$ times.	87
4.5	The normal distribution with mean = 0 and standard deviation = 1. The x-axis corresponds to the value of some variable, and the y-axis tells us something about how likely we are to observe that value. However, notice that the y-axis is labelled Probability Density and not Probability. There is a subtle and somewhat frustrating characteristic of continuous distributions that makes the y axis behave a bit oddly: the height of the curve here isn't actually the probability of observing a particular x value. On the other hand, it is true that the heights of the curve tells you which x values are more likely (the higher ones!).	90
4.6	Formula for the normal distribution	91
4.7	Simple random sampling without replacement from a finite population	96
4.8	Biased sampling without replacement from a finite populations.	97
4.9	Simple random sampling with replacement from a finite population.	98
4.10	The population distribution of IQ scores (panel a) and two samples drawn randomly from it. In panel b we have a sample of 100 observations, and panel c we have a sample of 10,000 observations.	102
4.11	A uniform distribution illustrating the probabilities of sampling the numbers 1 to 10. In a uniform distribution, all numbers have an equal probability of being sampled, so the line is flat indicating all numbers have the same probability	106
4.12	A histogram showing the sample means for 10,000 samples, each size 20, from the uniform distribution of numbers from 1 to 10. The expected mean is 5.5, and the histogram is centered on 5.5. The mean of each sample is not always 5.5 because of sampling error or chance	107
4.13	Each panel shows a histogram of a different sampling statistic	109
4.14	Comparison of two normal distributions, and histograms for the sampling distribution of the mean for different samples-sizes. The range of sampling distribution of the mean shrinks as sample-size increases	111
4.15	Illustration that the shape of the sampling distribution of the mean is normal, even when the samples come from a non-normal (uniform in this case) distribution	112
4.16	Illustration that the shape of the sampling distribution of the mean is normal, even when the samples come from an exponential distribution	113
4.17	A normal distribution. Each line represents a standard deviation from the mean. The labels show the proportions of scores that fall between each bar.	114
4.18	A normal distribution. Each line represents a standard deviation from the mean. The labels show the proportions of scores that fall between each bar.	115

4.19	The sampling distribution of the sample standard deviation for a two IQ scores experiment. The true population standard deviation is 15 (dashed line), but as you can see from the histogram, the vast majority of experiments will produce a much smaller sample standard deviation than this. On average, this experiment would produce a sample standard deviation of only 8.5, well below the true value! In other words, the sample standard deviation is a biased estimate of the population standard deviation.	125
4.20	An illustration of the fact that the sample mean is an unbiased estimator of the population mean.	126
4.21	An illustration of the fact that the the sample standard deviation is a biased estimator of the population standard deviation.	127
5.1	Uniform distribution showing that the numbers from 1 to 10 have an equal probability of being sampled	134
5.2	Histogram of 100 numbers randomly sampled from the uniform distribution containing the integers from 1 to 10	136
5.3	Histograms for 10 different samples from the uniform distribution. Each sample contains 20 numbers. The histograms all look quite different. The differences between the samples are due to sampling error or random chance.	137
5.4	Which of these two samples came from a uniform distribution?	138
5.5	Histograms for different samples from a uniform distribution. $N = 100$ for each sample.	139
5.6	Histograms for different samples from a uniform distribution. $N = 1000$ for each sample.	140
5.7	Histograms for different samples from a uniform distribution. $N = 100,000$ for each sample.	141
5.8	Which of these samples came from a uniform distribution?	142
5.9	Which of these samples came from a uniform distribution?	143
5.10	Counts of green gumballs picked randomly by each hand.	145
5.11	10 simulated replications of picking gumballs. Each replication gives a slightly different answer. Any difference between the bars is due to chance, or sampling error. This shows that chance alone can produce differences, just by the act of sampling.	147
5.12	A look at the differences between number of each kind of gumball for the different replications. The difference should be zero, but sampling error produces non-zero differences.	148
5.13	Replicating the experiment 100 times, and looking at the differences each time. There are many kinds of differences that chance alone can produce.	149
5.14	A histogram of the differences obtained by chance over 10,000 replications. The most frequency difference is 0, which is what we expect by chance. But the differences can be as large as -10 or +10. Larger differences occur less often by chance. Chance can't do everything.	149
5.15	Histogram of mean differences arising by chance.	154

5.16	Applying decision boundaries to the histogram of mean differences. The boundaries identify what differences chance did or did not produce in the simulation.	156
5.17	The question marks refer to an area where you have some uncertainty. Differences inside the question mark region do not happen very often by chance. When you find differences of these sizes, should you reject the idea that chance caused your difference? You will always have some uncertainty associated with this decision because it is clear that chance could have caused the difference. But, chance usually does not produce differences of these sizes.	158
5.18	The shading of the blue bars indicates levels of confidence in whether a difference could have been produced by chance. Darker bars represent increased confidence that the difference was not produced by chance. Bars get darker as the mean difference increases in absolute value.	159
5.19	The range or width of the differences produced by chance shrinks as sample-size increases.	162
5.20	A graph of the maximum and minimum mean differences produced by chance as a function of sample-size. The range narrows as sample-size increases showing that chance alone produces a smaller range of mean differences as sample-size increases.	163
5.21	The red line represents the size of a mean difference that a researcher may be interested in detecting. All of the dots outside (above or below) the red line represent designs with small sample-sizes. When a difference of 10 occurs for these designs, we can rule out chance with confidence. The dots between the red lines represent designs with larger sample-sizes. These designs never produce differences as large as 10, so when those differences occur, we can be confident chance did not produce them.	164
5.22	A histogram of simulated mean differences for a randomization test	169
5.23	Would you expect a mean difference represented by the blue line to occur more or less often by chance compared to the mean difference represented by the red line?	170
6.1	Comparison of four distributions essential for hypothesis testing: Null, True Parent, Inferred Parent, and Sampling Distribution of the Sample Mean.	175
6.2	Two-tailed distribution	176
6.3	Side-by-side comparison of one-tailed t-test scenarios: exploring if our sample comes from a distribution either below or above the null hypothesis.	177
6.4	True parent distribution & null distribution are the same	178
6.5	Here, we have a perfectly representative sample	179
6.6	Imperfect, but convenient	180
6.7	Type 1 Error	181
6.8	Type 2 Error	182
6.9	$H_a: < X$	184
6.10	$H_a: < X$	185
6.11	$H_a: < X$	186

6.12	Ha: X	187
6.13		188
6.14		189
6.15		190
6.16		191
6.17	Each panel shows hypothetical distributions for two conditions. As the effect-size increases, the difference between the distributions become larger.	196
6.18	Each panel shows hypothetical distributions for two conditions. As the effect-size increases, the difference between the distributions become larger.	197
6.19	This figure shows power as a function of effect-size (Cohen's d) for a between-subjects independent samples t-test, with N=10, and alpha criterion 0.05. . . .	202
6.20	This figure shows power as a function of N for a between-subjects independent samples t-test, with d=0.2, and alpha criterion 0.05.	203
6.21	Effect size as a function of p-values for type 1 Errors under the null, for a paired samples t-test.	207
6.22	Each panel shows a histogram of a different sampling statistic.	208
7.1	A scatter plot with sample mean on the x-axis, and standard error of the mean on the y-axis	214
7.2	A histogram of the sample means divided by the sample standard errors, this is a t-distribution.	214
7.3	Sampling distributions for the mean, standard deviation, standard error of the mean, and <i>t</i>	217
7.4	Applying decision criteria to the <i>t</i> -distribution. Histogram of <i>ts</i> from samples (n=20) drawn from the same normal distribution (u=0, sd=1)	219
7.5	A distribution of <i>t</i> -values that can occur by chance alone, when there is no difference between the sample and a population	225
7.6	Splitting the <i>t</i> distribution up into regions each containing 10% of the <i>t</i> -values. The width between the bars narrows as they approach the center of the distribution, where there are more <i>t</i> -values.	227
7.7	The critical value of <i>t</i> for an alpha criterion of 0.05. 5% of all <i>ts</i> are at this value or larger	229
7.8	A case where the observed value of <i>t</i> is much less than the critical value for a one-directional t-test.	230
7.9	Critical values for a two-tailed test. Each line represents the location where 2.5% of all <i>ts</i> are larger or smaller than critical value. The total for both tails is 5%	231
7.10	The width of the <i>t</i> distribution shrinks as sample size and degrees of freedom (from 4 to 100) increases.	234
7.11	The distribution of <i>t</i> -values under the null. These are the <i>t</i> values that are produced by chance alone.	240
7.12	The distribution of <i>p</i> -values that are observed is flat under the null.	240
7.13	Simulating <i>ts</i> in R.	241

7.14	Simulating ps in R.	241
7.15	1000 simulated ts from the null distribution	242
7.16	1000 simulated ps from the null distribution	243
7.17	1000 ts when there is a true difference	244
7.18	1000 ps when there is a true difference	244
7.19	1000 ts for $n = 100$, when there is a true effect	246
7.20	1000 ps for $n = 100$, when there is a true effect	246
7.21	1000 ts for $n = 100$, when there is a true effect	247
7.22	1000 ps for $n = 100$, when there is a true effect	248
8.1	A simulation of 10,000 experiments from a null distribution where there is no differences. The histogram shows 10,000 F -values, one for each simulation. These are values that F can take in this situation. All of these F -values were produced by random sampling error.	259
8.2	The critical value for F where 5% of all F -values lie beyond this point	260
8.3	Different patterns of group means under the null (all scores for each group sampled from the same distribution).	263
8.4	Different patterns of group means under the null (all scores for each group sampled from the same distribution).	264
8.5	Different patterns of group means under the null (sampled from same distribution) when F is less than 1.	265
8.6	Different patterns of group means under the null when F is above critical value (these are all type I Errors).	266
8.7	Mean number of intrusive memories per week as a function of experimental treatments.	268
9.1	Illustration showing how the total sums of squares are partitioned differently for a between versus repeated-measures design	276
9.2	Close-up showing that the Error term is split into two parts in the repeated measures design	277
9.3	Equations for computing the ANOVA table for a repeated measures design	278
9.4	Comparing critical F values for a between and repeated measures design	285
9.5	p -value distributions for a between and within-subjects ANOVA	287
9.6	p -value distribution for within-subjects design with $n = 30$	288
9.7	Results from Behmer & Crump (2017)	289
10.1	Structure of 2x2 factorial designs	292
10.2	Example data illustrating the influence of different pollution types on aquatic macroinvertebrate species diversity	294
10.3	Example data showing how the Pollution effect could be modulated by a Flow manipulation. Pollution plotted on the x-axis, makes it more difficult to compare the changes in the pollution effect between flow rate conditions	295

10.4	Example data showing how the pollution effect could be modulated by a flow rate manipulation. Flow condition plotted on the x-axis, makes it easier to compare the changes in the pollution effect between Flow conditions	296
10.5	same example means plotted using bar graphs or line graphs, and with Pollution or Flow on the x-axis	298
10.6	Each panel shows the mean for different effects in the design	299
11.1	8 Example patterns for means for each of the possible kinds of general outcomes in a 2x2 design.	317
11.2	Line graphs showing 8 possible general outcomes for a 2x2 design.	318
11.3	Example means showing a generally consistent main effect along with an interaction	319
11.4	Example data showing how an interaction exists, and a main effect does not, even though the means for the main effect may show a difference	320
11.5	Example means for a 3x2 factorial design in environmental science	323
11.6	Example means for a 3x2 design in environmental science with only one main effect	324
11.7	Example means for a 3x2 design in environmental science showing a different interaction pattern	325
11.8	Example means from a 2x2x2 design in environmental science with no three-way interaction.	326
11.9	Example means from a 2x2x2 design in environmental science with a three-way interaction.	326
13.1	Each panel shows hypothetical distributions for two conditions. As the effect-size increases, the difference between the distributions become larger.	348
13.2	Each panel shows hypothetical distributions for two conditions. As the effect-size increases, the difference between the distributions become larger.	349
13.3	This figure shows power as a function of effect-size (Cohen's d) for a between-subjects independent samples t-test, with N=10, and alpha criterion 0.05.	354
13.4	This figure shows power as a function of N for a between-subjects independent samples t-test, with d=0.2, and alpha criterion 0.05.	355
13.5	Effect size as a function of p-values for type 1 Errors under the null, for a paired samples t-test.	359
13.6	Each panel shows a histogram of a different sampling statistic.	360

Preface

Second Draft (version 0.1 = August 18th, 2025)

Welcome to the second edition of this Open Educational Resource (OER) textbook, specifically adapted for Environmental Science students enrolled in the SPEA E-538 statistics course at Indiana University (IU).

This textbook is an adaptation of a thorough introductory statistics textbook originally developed for undergraduate Psychology students by Matthew Crump and colleagues (refer to Acknowledgements for more details). As part of IU's Course Materials Fellowship Program (CMFP), I've had the opportunity to mold this material, refining it to serve as a specialized resource for students studying Environmental Science.

Online Textbook: https://malloryb.github.io/statistics_E538/

Citation for original textbook: Crump, M. J. C., Navarro, D. J., & Suzuki, J. (2019, June 5). Answering Questions with Data (Textbook): Introductory Statistics for Psychology Students. <https://doi.org/10.17605/OSF.IO/JZE52>

All resources are released under a creative commons licence [CC BY-SA 4.0](#). Click the link to read more about the license, or read more below in the license section.

Acknowledgements

I wish to express my deepest appreciation to the contributors of the original textbook, without whom this adaptation would not have been possible. I am deeply grateful for the expertise and vision of Matthew Crump, Alla Chavarga, Anjali Krishnan, Jeffrey Suzuki, and Stephen Volz. Their exceptional groundwork laid the foundation for this project.

My heartfelt thanks also go to the Course Materials Fellowship Program (CMFP) at Indiana University (IU), which has been instrumental in supporting this adaptation effort. The CMFP is an initiative designed to incentivize the discovery, implementation, and creation of cost-effective course materials. Its aim is to foster the use of Open Educational Resources (OERs)—freely accessible and customizable learning materials that make education more equitable and accessible.

I am particularly grateful to Sarah Hare, the Bloomington lead for the CMFP program, and Adam Mazel, a digital publishing librarian at IU. Their guidance, expertise, and steadfast

support have been crucial to this project's success. Their contributions to the advancement of affordable and accessible education are truly commendable.

CC BY-SA 4.0 license

This license means that you are free to:

- Share: copy and redistribute the material in any medium or format
- Adapt: remix, transform, and build upon the material for any purpose, even commercially.

The licensor cannot revoke these freedoms as long as you follow the license terms.

Under the following terms:

- Attribution: You must give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
- ShareAlike: If you remix, transform, or build upon the material, you must distribute your contributions under the same license as the original.
- No additional restrictions: You may not apply legal terms or technological measures that legally restrict others from doing anything the license permits.

Copying the textbook

This textbook was written in R-Studio, using R Markdown, and compiled into a web-book format using the bookdown package. In general, I thank the larger R community for all of the amazing tools they made, and for making those tools open, so that I could use them to make this thing.

All of the source code for compiling the book is available from the GitHub repository for the original textbook:

<https://github.com/CrumpLab/statistics>

and my github repository:

https://github.com/malloryb/statistics_E538

In principle, anyone can fork or download the E-538 textbook, or the original textbook, which is what I did. You can load the Rproj file in RStudio, compile the entire book, and then edit the individual .Rmd files for content and style to fit your needs.

If you'd like to contribute to this version, you can submit pull requests on GitHub or use the issues tab to share suggestions.

The vision behind this textbook

The aim of this textbook is twofold: to make core statistical concepts accessible to Environmental Science students, and to promote the use of open-source tools like R as flexible, transparent resources for learning and research.